

Notes

Assignments

◆ 5.23, 5.29, 5.33

Tutorial problems

◆ Property of DTFT: 5.24

◆ Difference equation of LTI systems: 5.36

DTFT

- Therefore, we get the discrete-time Fourier transform pair

Discrete-Time Fourier Transform

$$\text{Synthesis Equation: } x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

$$\text{Analysis Equation: } X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

- Observations
 - ▶ Continuous spectrum: Similar to CTFT
 - ▶ Periodic with period 2π : Different from CTFT
 - ▶ Low frequency: close to 0 and 2π ; high frequency: close to π

Discrete-Time Fourier Transform

$$\text{Synthesis Equation: } x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

$$\text{Analysis Equation: } X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

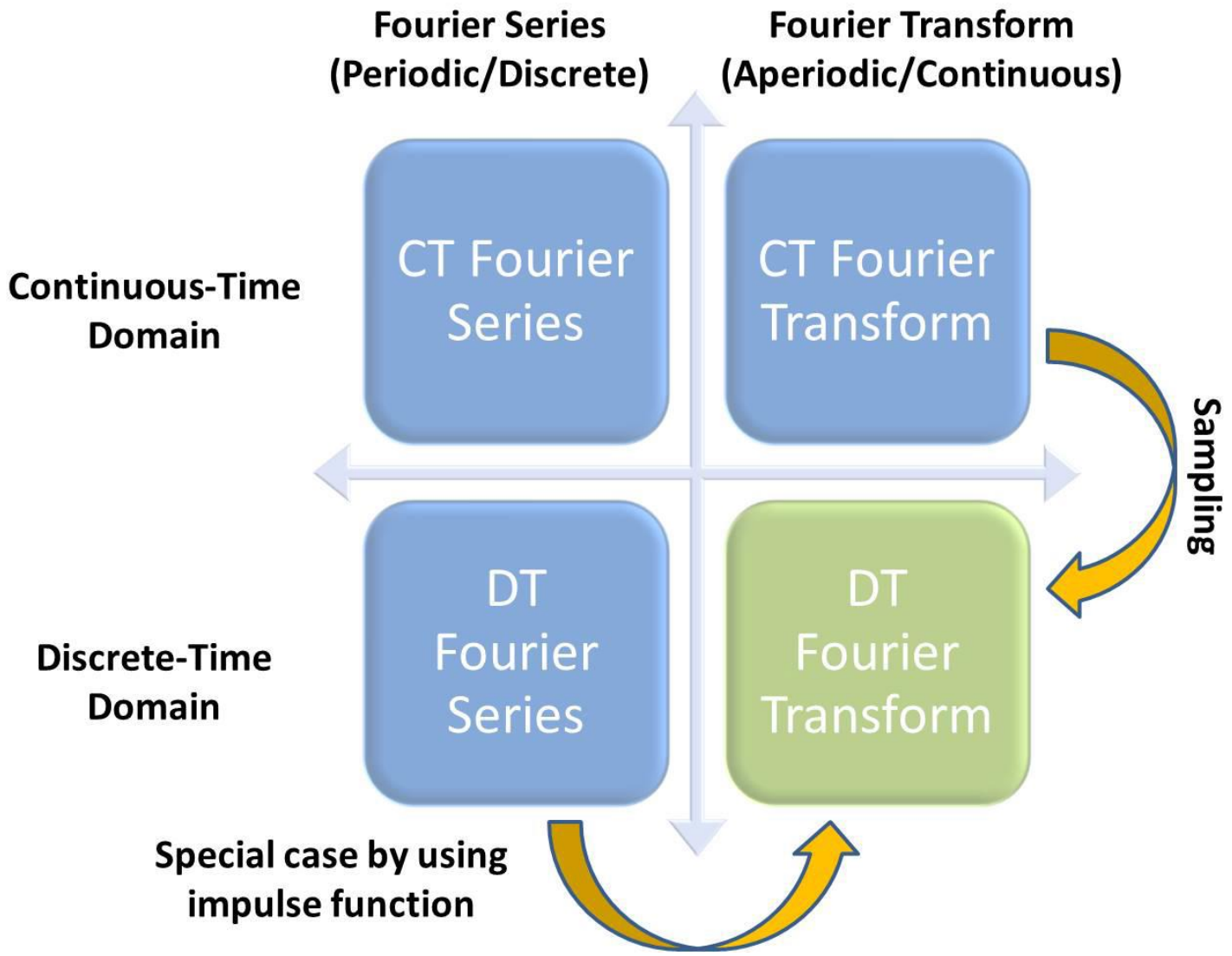
DT Fourier Series Pair $\left(\omega_o = \frac{2\pi}{N}\right)$

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_o n} \quad (\text{Synthesis equation})$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_o n} \quad (\text{Analysis equation})$$

Frequency domain

Time domain



- If signal x is **periodical**, then X is computed by Fourier _____, and its shape is _____?

Further more,

- ◆ If signal x is CT, then a_k is _____?

- ◆ If signal x is DT, then a_k is _____?

- If signal x is **aperiodical**, then X is computed by Fourier _____, and its shape is _____?

Further more,

- ◆ If signal x is CT, then X is _____?

- ◆ If signal x is DT, then X is _____?

DTFT with Periodic Signals (1/2)

Fourier Transform of $e^{j\omega n}$

The following transform pair is actually NOT rigorously defined:

$$x[n] = e^{j\omega_0 n} \longleftrightarrow X(e^{j\omega}) = \sum_{l=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 - 2\pi l)$$

- Synthesis:

$$\frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{2\pi} \sum_{l=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 - 2\pi l) e^{j\omega n} d\omega = e^{j\omega_0 n}$$

- Analysis:

$$\sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} e^{j(\omega_0 - \omega)n} \quad \text{converge??}$$

DTFT with Periodic Signals (2/2)

- According to the Fourier series, for a periodic signal with period N :

$$x[n] = a_0 + a_1 e^{j(2\pi/N)n} + \dots + a_k e^{jk(2\pi/N)n} + \dots + a_{N-1} e^{j(N-1)(2\pi/N)n}$$

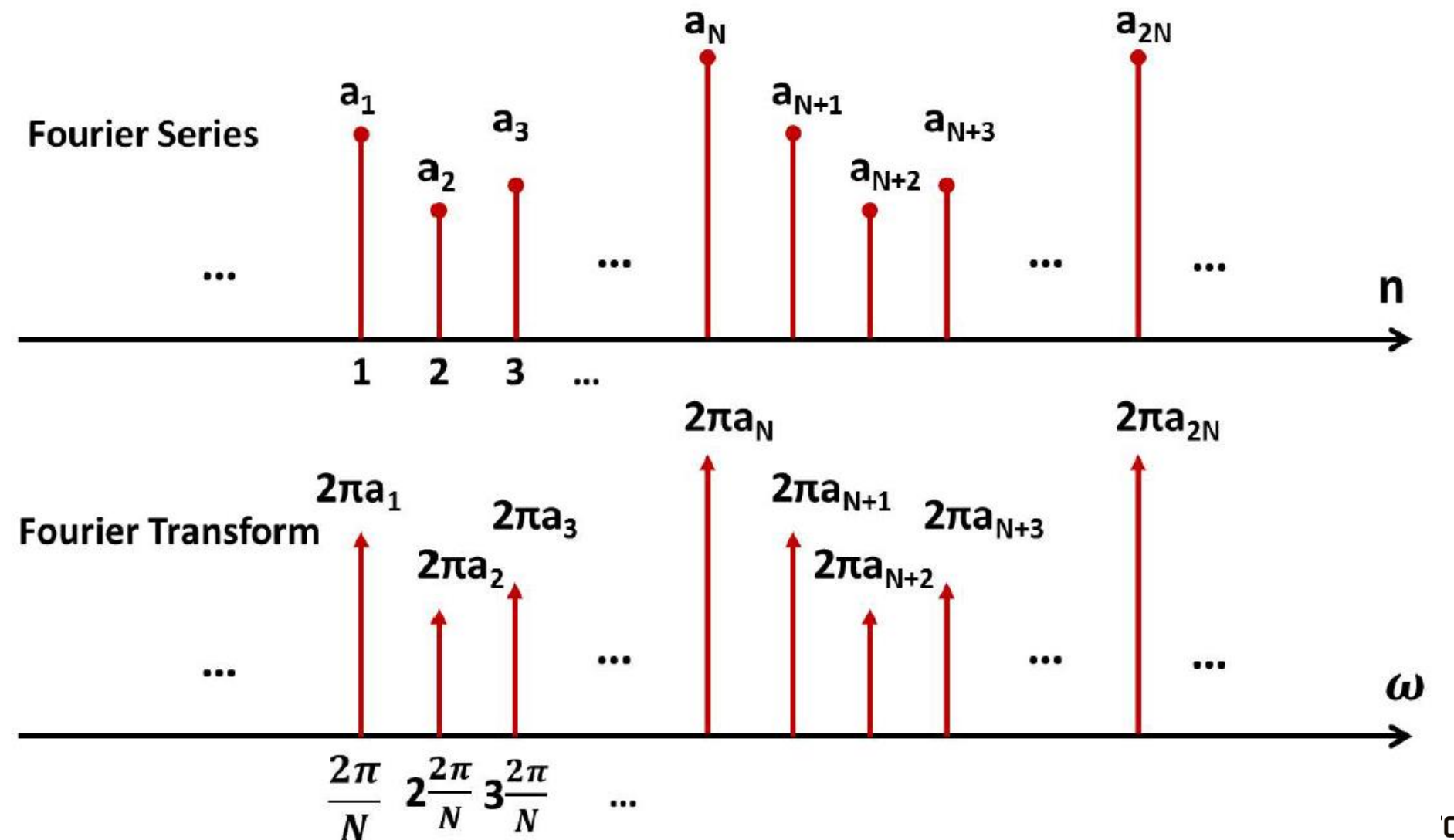
- $e^{jk(2\pi/N)n} \longleftrightarrow \sum_{l=-\infty}^{\infty} 2\pi \delta(\omega - k(2\pi/N) - 2\pi l)$

- Then, due to the linearity of Fourier transform

$$\begin{aligned} \mathcal{F}\{x[n]\} &= \sum_{k=0}^{N-1} a_k \mathcal{F}\{e^{jk(2\pi/N)n}\} = \sum_{k=0}^{N-1} a_k \sum_{l=-\infty}^{\infty} 2\pi \delta(\omega - k(2\pi/N) - 2\pi l) \\ &= \sum_{l=-\infty}^{\infty} \sum_{k=0}^{N-1} a_k 2\pi \delta(\omega - k(2\pi/N) - 2\pi l) \end{aligned}$$

- Fourier transform of a periodic signal is a periodic sequence of impulses
 - ▶ What's the period? How many impulses within one period?

Fourier Series v.s. Fourier Transform



Example: Discrete-Time Impulse Chain

- What's the Fourier transform of $x[n] = \sum_{k=-\infty}^{+\infty} \delta[n - kN]$?
- First of all, we calculate the Fourier series:

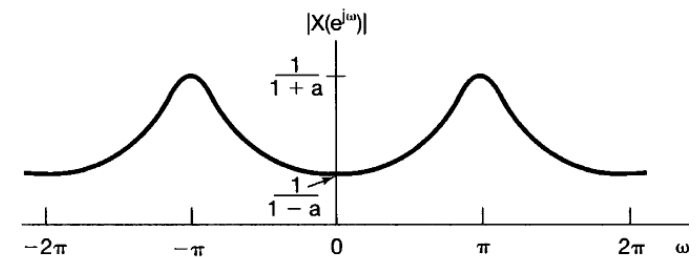
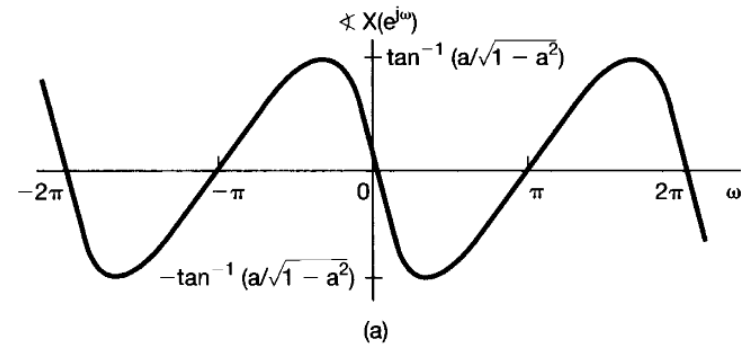
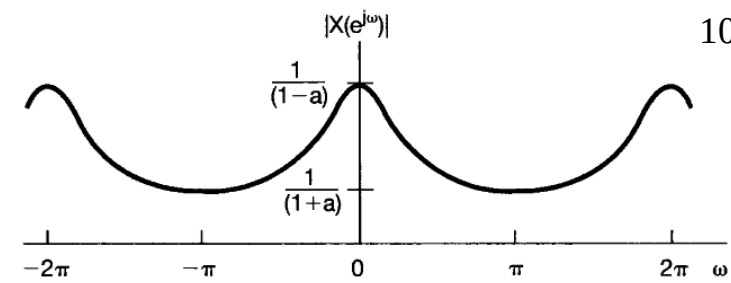
$$\begin{aligned} a_k &= \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk \frac{2\pi}{N} n} \\ &= \frac{1}{N} \sum_{n=\langle N \rangle} \sum_{k=-\infty}^{+\infty} \delta[n - kN] e^{-jk \frac{2\pi}{N} n} \\ &= \frac{1}{N} \sum_{n=\langle N \rangle} \delta[n] e^{-jk \frac{2\pi}{N} n} = \frac{1}{N} \end{aligned}$$

Example 5.1

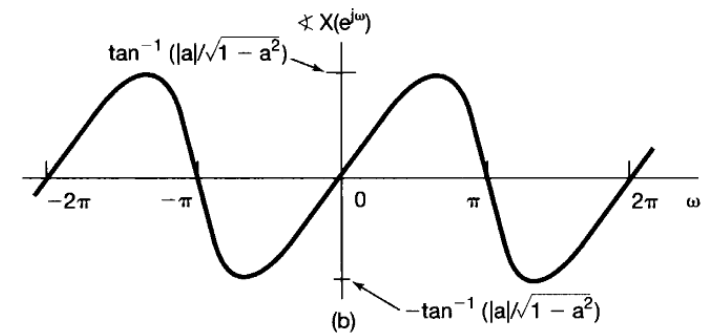
$$x[n] = a^n u[n], \quad |a| < 1.$$

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{+\infty} a^n u[n] e^{-j\omega n} \\ &= \sum_{n=0}^{\infty} (ae^{-j\omega})^n = \frac{1}{1 - ae^{-j\omega}}. \end{aligned}$$

$a > 0$

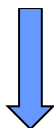


$a < 0$



Example 5.2

$$x[n] = a^{|n|}, \quad |a| < 1.$$



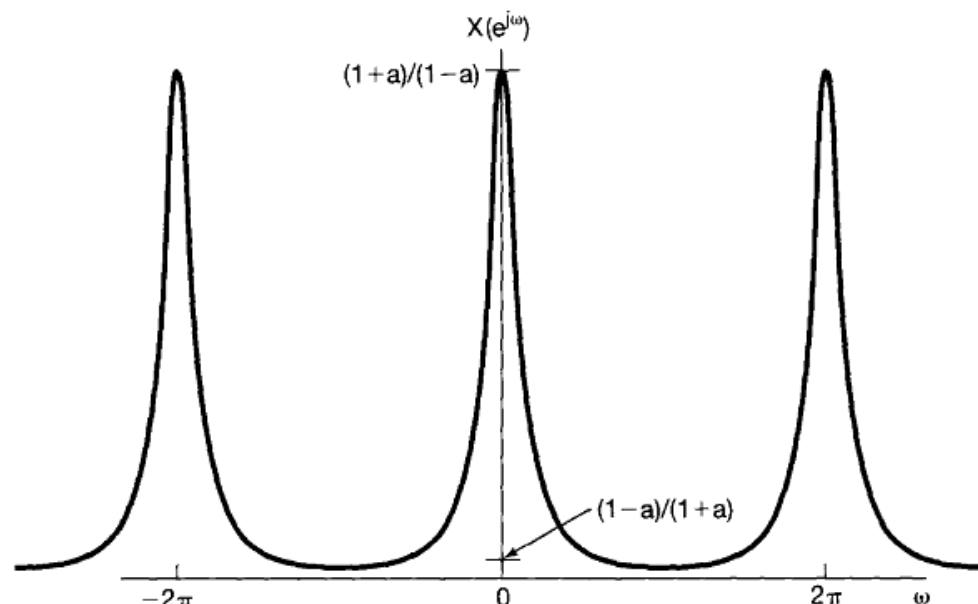
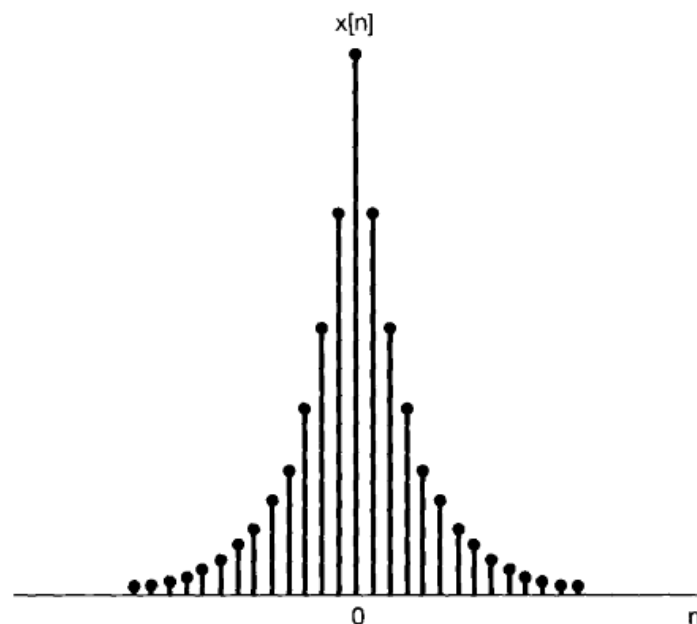
$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{+\infty} a^{|n|} e^{-j\omega n} \\ &= \sum_{n=0}^{\infty} a^n e^{-j\omega n} + \sum_{n=-\infty}^{-1} a^{-n} e^{-j\omega n}. \end{aligned}$$

$m = -n$

$$= \sum_{n=0}^{\infty} (ae^{-j\omega})^n + \sum_{m=1}^{\infty} (ae^{j\omega})^m.$$

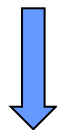
$$= \frac{1}{1 - ae^{-j\omega}} + \frac{ae^{j\omega}}{1 - ae^{j\omega}}$$

$$= \frac{1 - a^2}{1 - 2a \cos \omega + a^2}.$$



Example 5.4

$$x[n] = \delta[n].$$



$$X(e^{j\omega}) = 1$$

Periodicity, Linearity and Shifting

- Periodicity

$$X(e^{j(\omega+2\pi)}) = X(e^{j\omega})$$

- ▶ How about CTFT? Why?

- Linearity

$$ax_1[n] + bx_2[n] \longleftrightarrow aX_1(e^{j\omega}) + bX_2(e^{j\omega})$$

- Time Shifting and Frequency Shifting

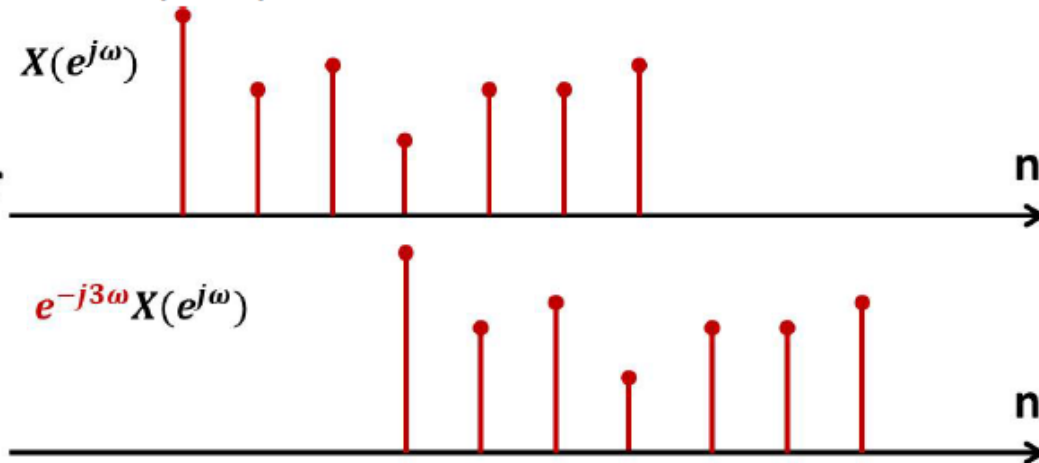
$$\begin{aligned} x[n - n_0] &\longleftrightarrow e^{-j\omega n_0} X(e^{j\omega}) \\ e^{j\omega_0 n} x[n] &\longleftrightarrow X(e^{j(\omega - \omega_0)}) \end{aligned}$$

Duality!

Illustration on Shifting

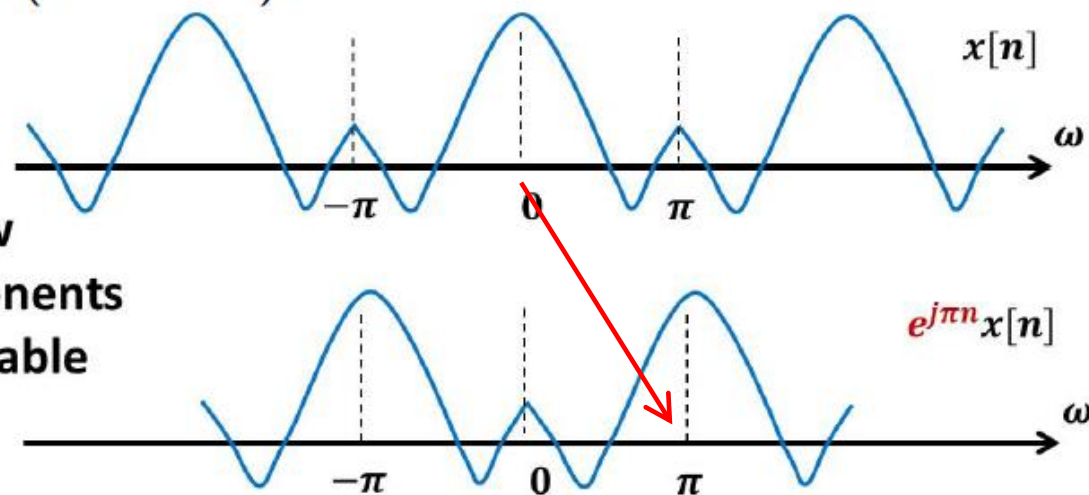
- $x[n - n_0] \longleftrightarrow e^{-j\omega n_0} X(e^{j\omega})$

Delay raises linear
phase shifting



- $e^{j\omega_0 n} x[n] \longleftrightarrow X(e^{j(\omega - \omega_0)})$

High and low
frequency components
are interchangeable



Conjugation, Differencing and Accumulation

- Conjugation

$$x^*[n] \longleftrightarrow X^*(e^{-j\omega})$$

- ▶ $X(e^{j\omega}) = X^*(e^{-j\omega}) \Leftrightarrow x[n]$ is real
- ▶ If $x[n]$ is real, then $\Re\{X(e^{j\omega})\}$ is even, $\Im\{X(e^{j\omega})\}$ is odd

- Differencing

$$x[n] - x[n-1] \longleftrightarrow (1 - e^{-j\omega})X(e^{j\omega})$$

- ▶ High-pass or low-pass?

- Accumulation

$$\sum_{m=-\infty}^n x[m] \longleftrightarrow \frac{1}{1 - e^{-j\omega}} X(e^{j\omega}) + \pi X(e^{j0}) \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)$$

- ▶ How to derive it via differencing?

Effect of Differencing



- $$J(i,j) = |M(i,j) - M(i+1,j+1)| + |M(i+1,j) - M(i,j+1)|$$

Fourier Transform of $u[n]$

- How to derive the Fourier transform of $u[n]$?
- Option 1: From definition of Fourier transform

$$\mathcal{F}\{u[n]\} = \sum_{n=-\infty}^{+\infty} u[n]e^{-j\omega n} = \sum_{n=0}^{+\infty} e^{-j\omega n} \quad \text{Converge??}$$

- Option 2: Since $u[n] = \sum_{m=-\infty}^n \delta[m]$, according to the property of accumulation

$$\mathcal{F}\{u[n]\} = \frac{1}{1 - e^{-j\omega}} + \pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)$$

- Observation: Fourier transform of $u[n]$ does not exist according to the definition; however, it can be expressed in terms of $\delta(\cdot)$

Time Reversal and Expansion

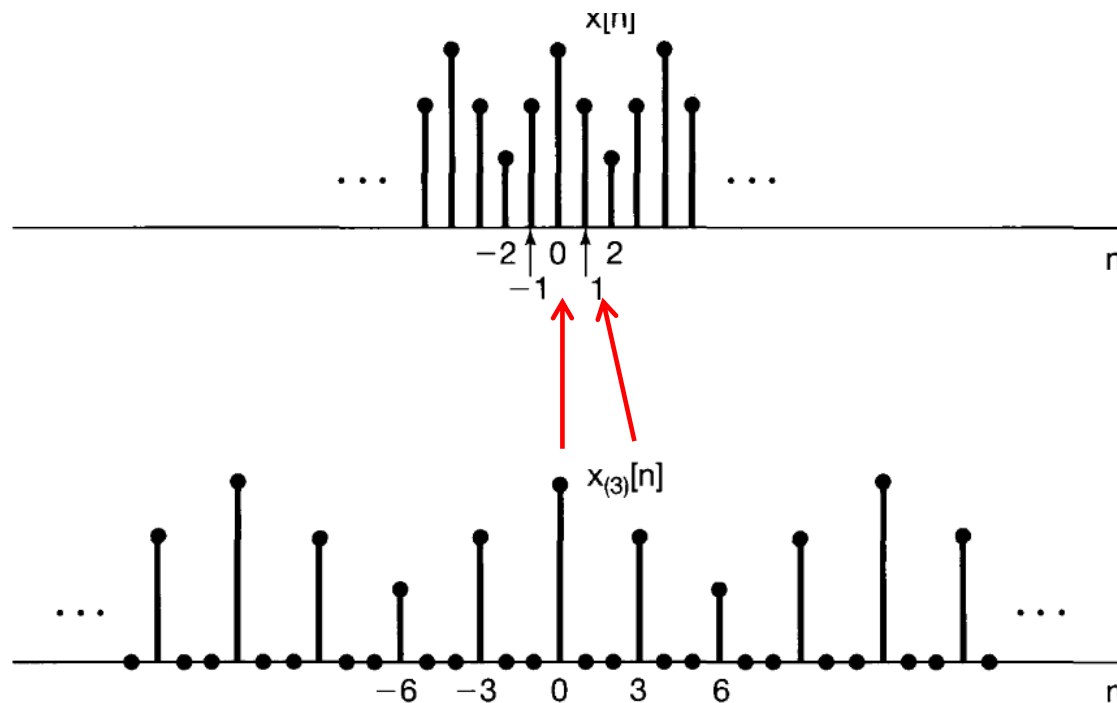
- Time Reversal

$$x[-n] \longleftrightarrow X(e^{-j\omega})$$

- Time Expansion

► Define $x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n \text{ is a multiple of } k \\ 0, & \text{Otherwise} \end{cases}$, then

$$x_{(k)}[n] \longleftrightarrow X(e^{jk\omega})$$



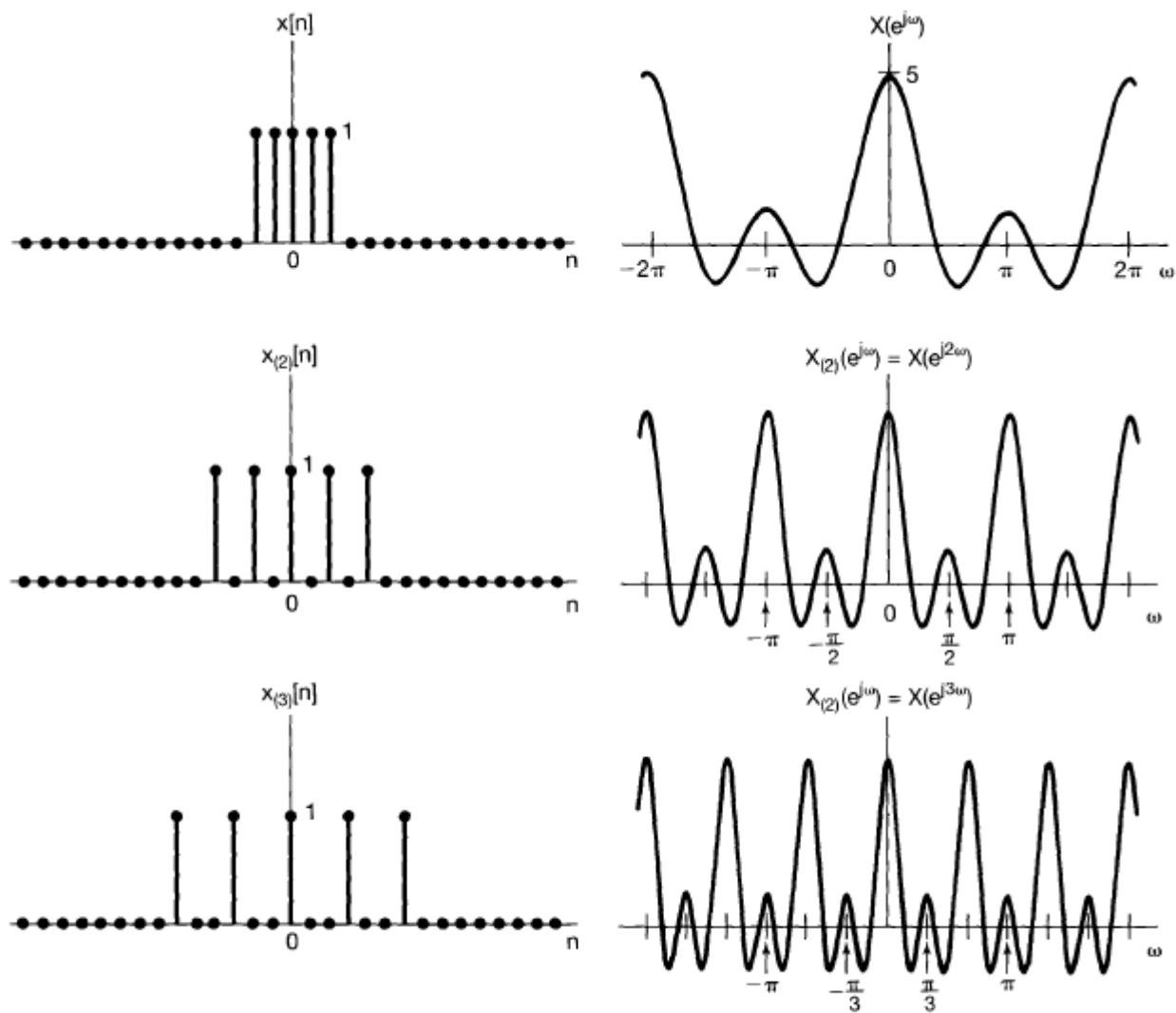


Figure 5.14 Inverse relationship between the time and frequency domains: As k increases, $x_{(k)}[n]$ spreads out while its transform is compressed.

Differentiation and Parseval

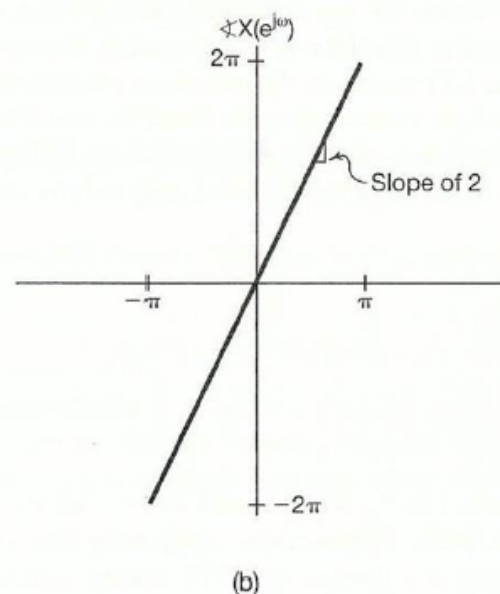
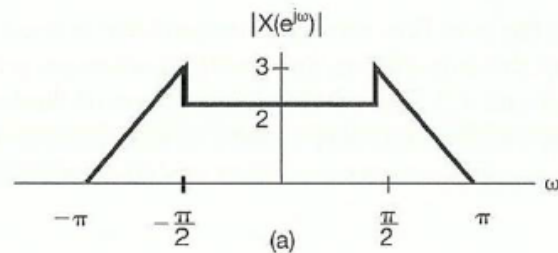
- Differentiation in Frequency

$$nX[n] \longleftrightarrow j \frac{dX(e^{j\omega})}{d\omega}$$

- Parseval's Relation

$$\sum_{n=-\infty}^{+\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{2\pi} |X(e^{j\omega})|^2 d\omega$$

► Energy density spectrum — $\frac{|X(e^{j\omega})|^2}{2\pi}$



- See textbook, Example 5.10
- Spectrum within $[-\pi, \pi]$
- Is it periodic, real, even, of finite energy?

Convolution Property & LTI Systems (1/2)

- Let $h[n]$ be the **impulse response** of certain LTI system
- The output $y[n]$ of input $x[n]$ is given by $y[n] = x[n] * h[n]$
- For input signal $x[n] = e^{j\omega n}$,

$$y[n] = \sum_{k=-\infty}^{\infty} h[k] e^{j\omega(n-k)} = e^{j\omega n} \underbrace{\sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k}}_{\text{Frequency Response } H(e^{j\omega})}$$

- For periodic input signal $x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(\frac{2\pi}{N})n}$:

$$y[n] = \sum_{k=\langle N \rangle} a_k H(e^{jk(\frac{2\pi}{N})}) e^{jk(\frac{2\pi}{N})n}$$

$$b_k = a_k H(e^{jk(\frac{2\pi}{N})})$$

Convolution Property

If $y[n] = x_1[n] * x_2[n]$, then

$$Y(e^{j\omega}) = X_1(e^{j\omega})X_2(e^{j\omega})$$

- For general input signal $x[n]$:

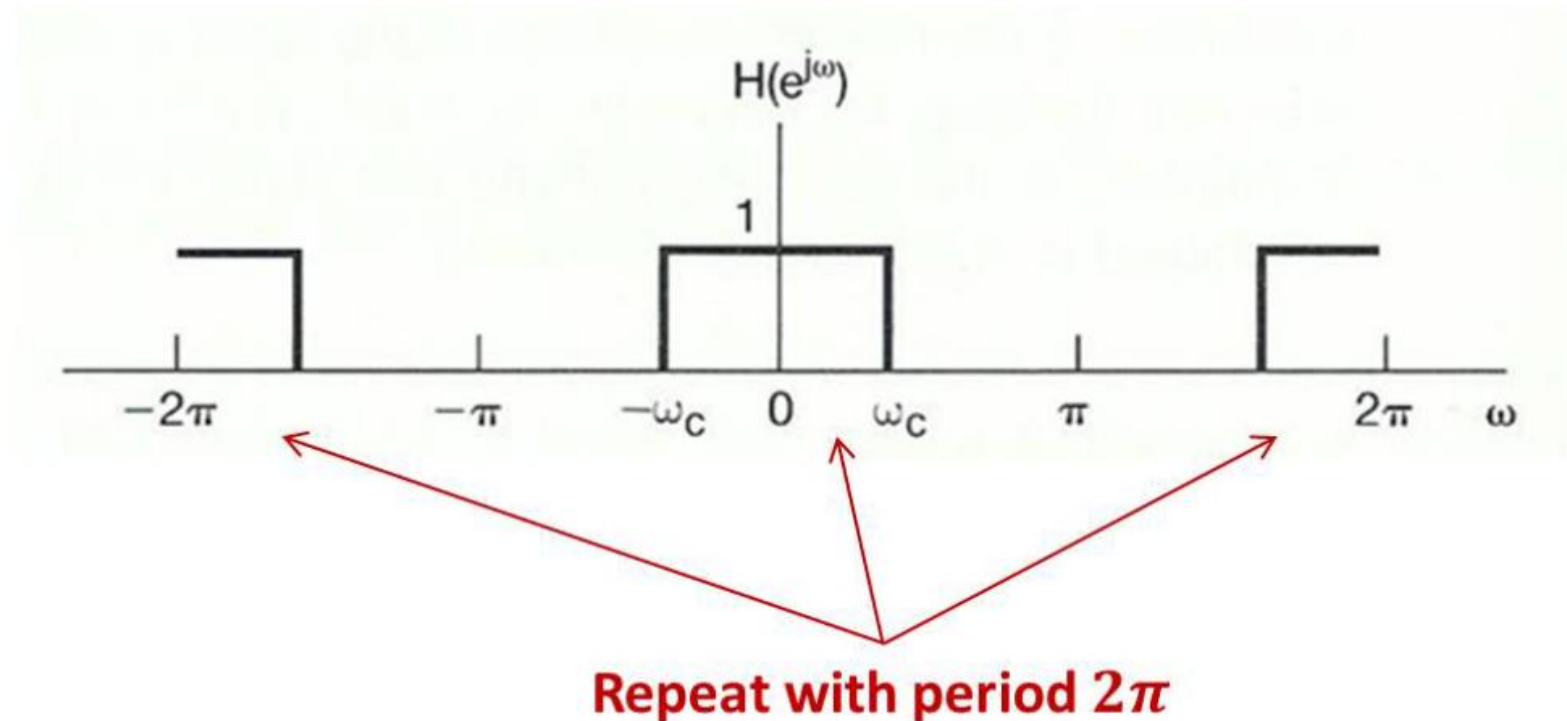
$$y[n] = x[n] * h[n] \longleftrightarrow Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$

- **Observation:** It's easier to evaluate LTI systems in frequency domain
- **Drawback:** Not every LTI system has frequency response
 - ▶ $h[n] = a^n u[n]$ ($a > 1$)
 - ▶ Stable LTI system has frequency response, because

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

Example: Ideal Low-Pass Filter (1/2)

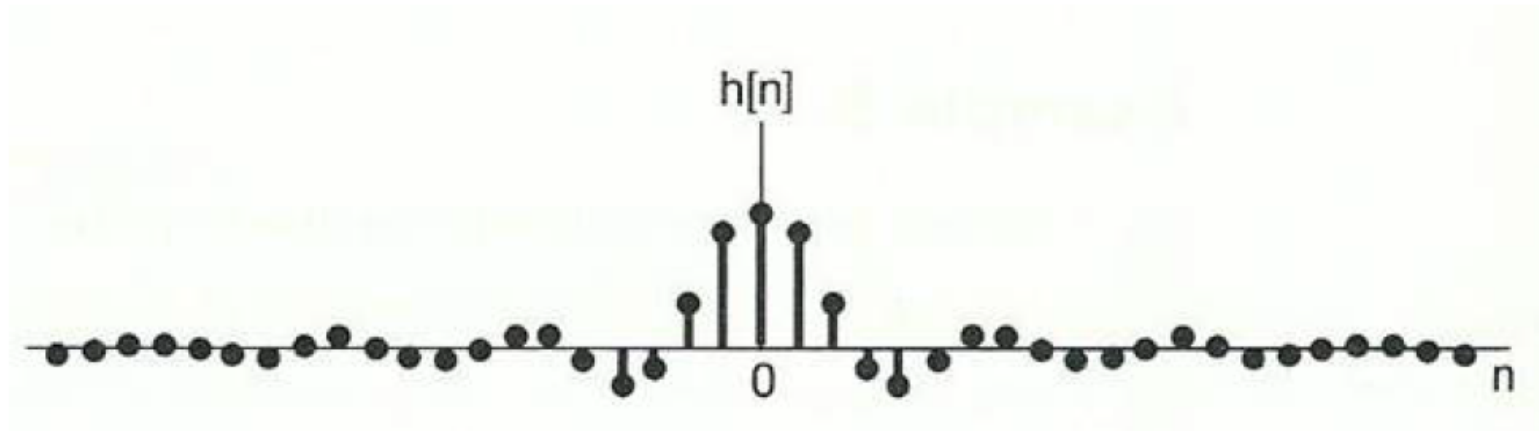
- What's ideal low-pass filter?
- Perfectly maintain the low-frequency component
- Perfectly cancel the high-frequency component



Example: Ideal Low-Pass Filter (2/2)

- Impulse response:

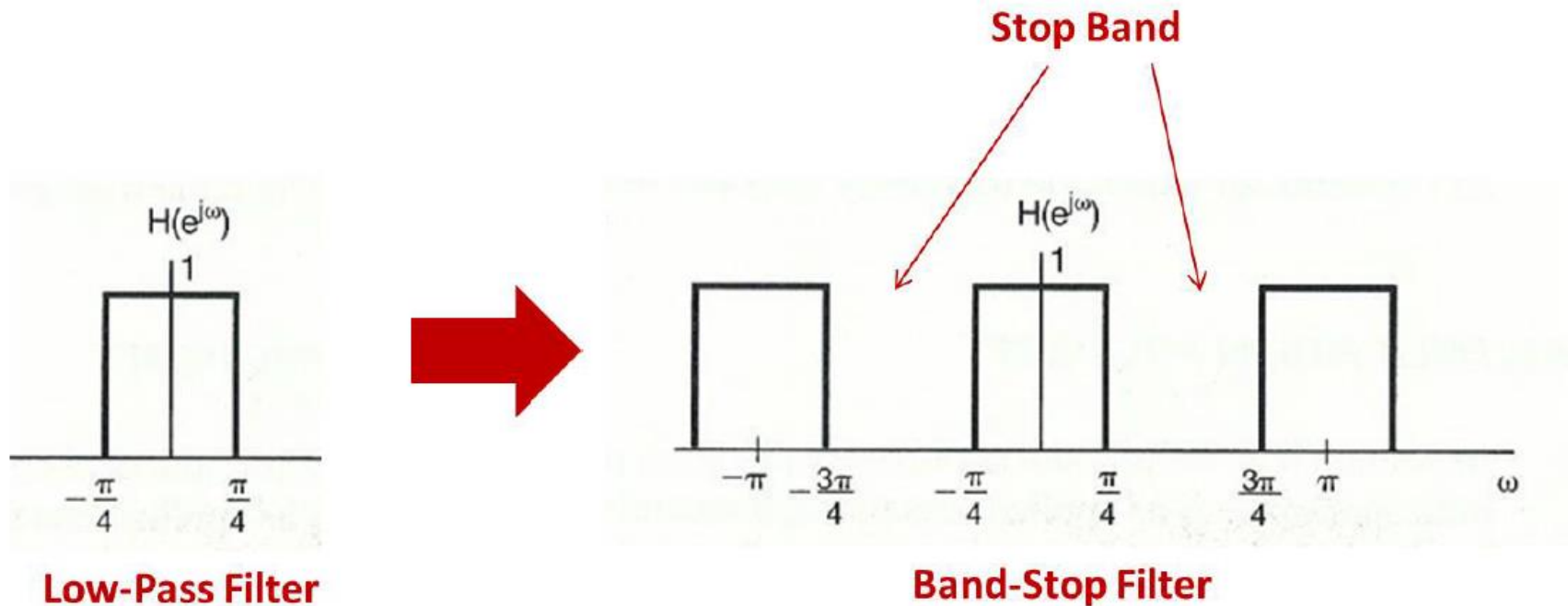
$$h[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega n} d\omega = \frac{\sin \omega_c n}{\pi n}$$



- Pros: no distortion in frequency domain
- Cons: non-causal
- See textbook, Example 5.12

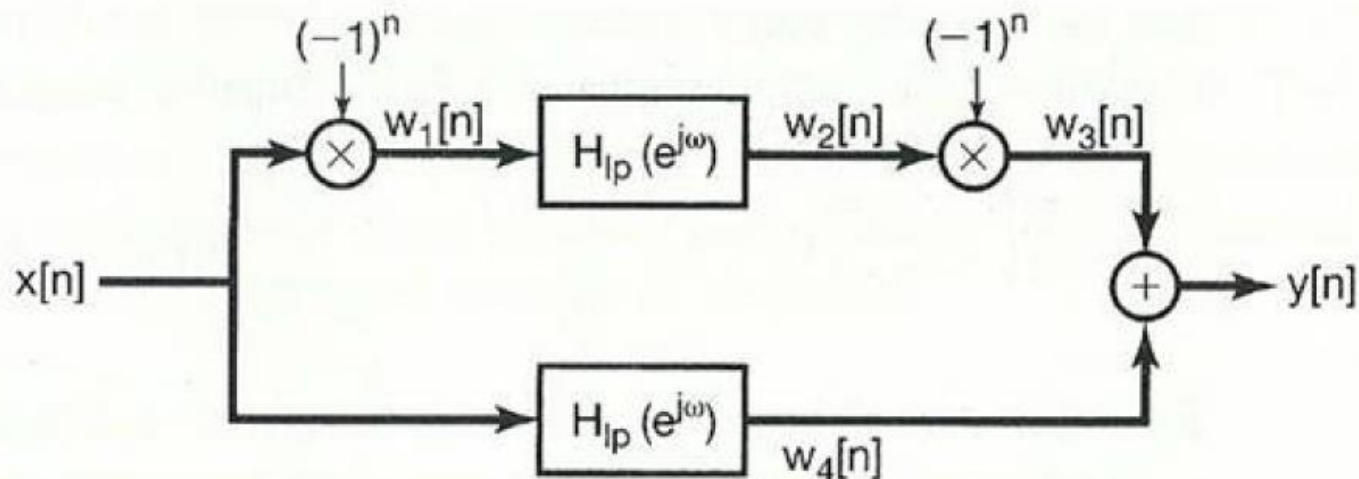
Example: Band-Stop Filter (1/2)

- Given low-pass filter $H_{lp}(e^{j\omega})$, how to generate band-stop effect from low-pass filter?



Example: Band-Stop Filter (2/2)

- Two branches: low-pass + high-pass
- See textbook, Example 5.14



- $(-1)^n = e^{j\pi n} \Rightarrow W_1(e^{j\omega}) = X(e^{j(\omega-\pi)})$
 $\Rightarrow W_2(e^{j\omega}) = H_{lp}(e^{j\omega})X(e^{j(\omega-\pi)})$
 $\Rightarrow W_3(e^{j\omega}) = H_{lp}(e^{j(\omega-\pi)})X(e^{j(\omega-2\pi)}) = H_{lp}(e^{j(\omega-\pi)})X(e^{j\omega})$

Multiplication Property

Let $y[n] = x_1[n]x_2[n]$, then

$$Y(e^{j\omega}) = \frac{1}{2\pi} \int_{2\pi} X_1(e^{j\theta}) X_2(e^{j(\omega-\theta)}) d\theta,$$

which is *periodic convolution* of $X_1(e^{j\omega})$ and $X_2(e^{j\omega})$

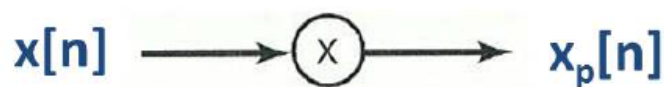
- Comparison: Multiplication of periodic signals

$$x_1[n] \longleftrightarrow a_k \quad \text{and} \quad x_2[n] \longleftrightarrow b_k$$

$$\Rightarrow x_1[n]x_2[n] \longleftrightarrow \sum_{k=\langle N \rangle} a_k b_{n-k} \quad \text{discrete-time periodic convolution}$$

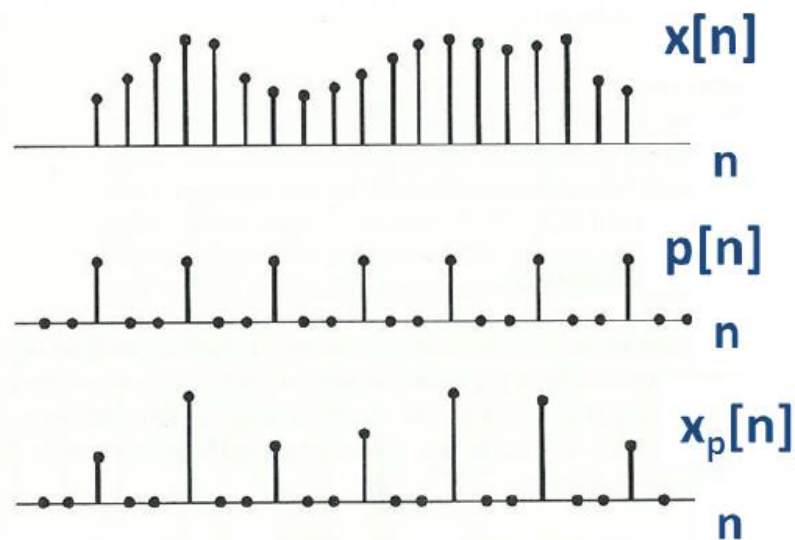
Example: Sampling on Discrete-Time Signals

- In digital signal processing, we may sample the discrete-time signals
- Example: image, audio, video compression
- Its mathematical model is give below:



$$p[n] = \sum_{k=-\infty}^{+\infty} \delta[n - kN]$$

$$x_p[n] = \sum_{k=-\infty}^{+\infty} x[kN] \delta[n - kN]$$



Example: Discrete-Time Impulse Chain

- What's the Fourier transform of $x[n] = \sum_{k=-\infty}^{\infty} \delta[n - kN]$?
- First of all, we calculate the Fourier series:

$$\begin{aligned} a_k &= \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk \frac{2\pi}{N} n} \\ &= \frac{1}{N} \sum_{n=\langle N \rangle} \sum_{k=-\infty}^{+\infty} \delta[n - kN] e^{-jk \frac{2\pi}{N} n} \\ &= \frac{1}{N} \sum_{n=\langle N \rangle} \delta[n] e^{-jk \frac{2\pi}{N} n} = \frac{1}{N} \end{aligned}$$

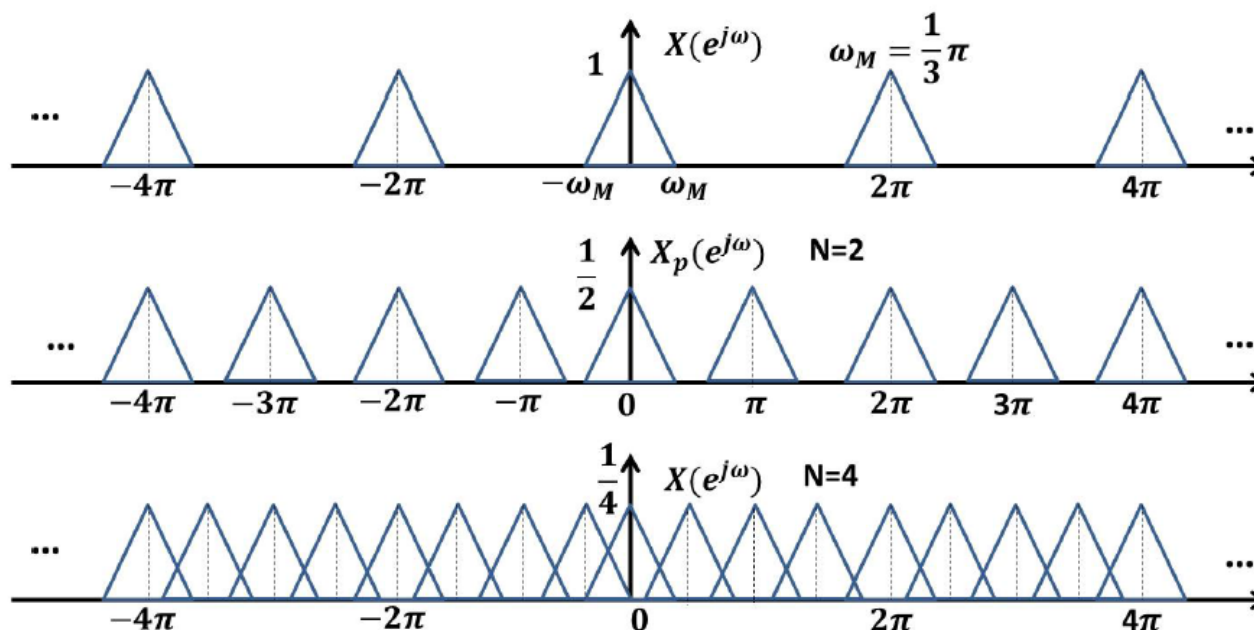
- Then, we have

$$X(e^{j\omega}) = \sum_{l=-\infty}^{\infty} \sum_{k=0}^{N-1} \frac{2\pi}{N} \delta(\omega - k(2\pi/N) - 2\pi l) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta(\omega - \frac{2\pi k}{N})$$

Example: Frequency Analysis

$$P(e^{j\omega}) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s) \quad \text{where } \omega_s = 2\pi/N$$

$$X_p(e^{j\omega}) = \frac{1}{2\pi} \int_{2\pi} P(e^{j\theta}) X(e^{j(\omega-\theta)}) d\theta = \frac{1}{N} \sum_{k=0}^{N-1} X(e^{j(\omega-k\omega_s)})$$



Example #8: LTI Systems Described by LCCDE's (Linear-constant-coefficient differential equations)

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}$$

Using the Differentiation Property

$$\frac{d^k x(t)}{dt^k} \longleftrightarrow (j\omega)^k X(j\omega)$$

$\Downarrow$ Transform both sides of the equation

$$\sum_{k=0}^N a_k (j\omega)^k Y(j\omega) = \sum_{k=0}^M b_k (j\omega)^k X(j\omega)$$

$\Downarrow$

$$Y(j\omega) = \underbrace{\left[\frac{\sum_{k=0}^M b_k (j\omega)^k}{\sum_{k=0}^N a_k (j\omega)^k} \right]}_{H(j\omega)} X(j\omega)$$

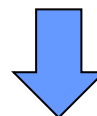
$$H(j\omega) = \left[\frac{\sum_{k=0}^M b_k (j\omega)^k}{\sum_{k=0}^N a_k (j\omega)^k} \right]$$

Linear constant-coefficient difference equations (LCCDE)

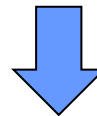
$$nx[n] \xleftrightarrow{\mathfrak{F}} j \frac{dX(e^{j\omega})}{d\omega}$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})}$$

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k].$$



$$\sum_{k=0}^N a_k e^{-jk\omega} Y(e^{j\omega}) = \sum_{k=0}^M b_k e^{-jk\omega} X(e^{j\omega})$$



$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\sum_{k=0}^M b_k e^{-jk\omega}}{\sum_{k=0}^N a_k e^{-jk\omega}}.$$

Example 5.19

Consider a causal LTI system that is characterized by the difference equation

$$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = 2x[n].$$

$$H(e^{j\omega}) = \frac{2}{1 - \frac{3}{4}e^{-j\omega} + \frac{1}{8}e^{-j2\omega}}.$$

$$H(e^{j\omega}) = \frac{2}{(1 - \frac{1}{2}e^{-j\omega})(1 - \frac{1}{4}e^{-j\omega})} = \frac{4}{1 - \frac{1}{2}e^{-j\omega}} - \frac{2}{1 - \frac{1}{4}e^{-j\omega}}$$

$$h[n] = 4\left(\frac{1}{2}\right)^n u[n] - 2\left(\frac{1}{4}\right)^n u[n].$$